



Durability Analysis - Component Rebuild Parts Cost

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

Customers in the mining industry focus on lowering their O&O (Owning and Operating) or TCO (Total Cost of Ownership) to maximize profit margin from the commodity they produce. Often when selecting equipment, one of the key criteria is lowest TCO for that class of equipment. Dealers would often develop the TCO estimation by 'localized' Caterpillar Global LCC (Life Cycle Cost) Centreline files based on customer's specific application needs. The Global Centreline files were developed during the NPI process when the model was released. Depending on the application and site maintenance strategy, the estimated component lives in the files might not line up with the actual components hours achieved in the field. The actual hours achieved by the components might deviate from target life estimated by the original localized file and at the same time well outside of the warranty period so customers would usually treat them as early PCR (Planned Component Replacement). When the rebuilder (Dealer CRC or 3rd Party) receives the component for rebuild, SIMS tickets might not be submitted. Hence, Caterpillar is not made aware of the life limiting part(s) for component(s). The purpose of this Best Practice is to demonstrate how Dealers can conduct piece parts consumption during rebuild and compare that against the localized builder file. Through the analysis, dealer can identify the piece parts that contribute to TCO misalignment and take appropriate actions accordingly.

2.0 Best Practice Description

The objectives of this Best Practice are:

- Conduct component rebuild parts cost analysis and its impact to TCO
- Provide feedback loop on component actual achieved life vs. estimated component life for Global LCC Centreline
- Provide feedback loop on component actual achieved life vs. estimated component life for Dealer 'localized' LCC Centreline
- Comparison of actual rebuild costs vs. estimated rebuild cost
- Identify misaligned parts consumption during PCR and initiate further investigation

This document provides process guidelines to establish a framework that allow the review of parts costs during each individual rebuild done by Dealers' Component Rebuild Centers.

3.0 Implementation Steps

- **Capture Parts Consumption Data for Every Rebuild Work Order** - The following data would need to be captured for every component rebuild by the Dealer Component Rebuild Center
 - Invoice/service call number
 - Part numbers and Group number – the part number and group number needs to be captured because there are some parts can be used on different components and different machines. Using both part number and group number to identify a part will provide the part with unique identity.
 - Parts quantity
 - Component number
 - Serial number
 - Component life achieved

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- **Construction of Component Analysis** – Dealers can conduct the analysis based on their business operation system and resource available to them. The steps below were based on using MS Excel as analysis tool and manual entering each analysis.
 - **Identify Relevant Piece Parts within Rebuild Component** - Obtain the electronics Parts Manual for the model to be analyzed. Filter out the piece parts related to the component being rebuild and analyzed. There two different ways to filter piece parts that belong to a component.
 - Components that do not have a unique serial number prefix such as differential and wheel ends on LMT. The analyst will need to identify all the parent/group number up to the piece part level in the component
 - Components that do have a unique serial number prefix such as engine and transmission on LMT, the analyst can filter it down by their respective serial number prefix from the parts manual
 - **Consideration for Machine Serial Number Break** – The parts consist can be filtered into different machine serial number breaks. It is important ensure the machine serial number break are selected with the corresponding machine serial number being analyzed
 - **Consideration for Part Number Change** – As improvements are made to the machine, new part numbers would be assigned to reflect the improvement. To take this into consideration, the analysis must include part number replacement dataset from the Dealer Parts System. It is a snapshot in time that shows prior, current and future piece part number at the time the file is obtained. The file can be sourced from Caterpillar Numerical Parts Record (NPR)
 - **Consideration for Part Numbers within Kits** – Kits are often used in rebuilds. The part numbers within a kit must be broken down to understand the piece parts consumption in rebuilds
 - **Create Component Consist for a Particular Machine Serial Number** – Once the above steps are completed, you would end up with the component parts consist that is reflective of a particular machine serial number.
 - **Obtain Component Rebuild Consist** – In this step you will need to obtain the actual rebuild consist that composed of the parts consumed for the rebuild for the same serial number as the prior step

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- **Conduct Component Analysis** – To conduct the component analysis, you will need to merge the data in the Component Consist with the Rebuild Consist in the following additional columns. If you are using MS Excel, merging the two consists would involve using of the Vlookup function.
 - Alternative Part Number
 - Previous Part Number
 - Replacing Part Number
 - Part Number Purchased in a Kit
 - Work Order Quantity – This is the average quantity of a particular part number replaced per rebuild work order
 - Work Order Average Replacement % - This is the work order quantity divide by the quantity needed in the component.
 - Dealer Standard Job Replacement % – Dealer would use the standard job replacement % to estimate and quote the rebuild job
 - Builder file replacement % - This information can be extracted from the Builder file
 - Delta % – This delta is the % difference between work order average replacement % vs. Builder file replacement %.
 - Dealer Net (D/N) Cost
 - Actual Cost – The product of D/N cost, Quantity and Work Order Average Replacement %
 - Builder Cost - The product of D/N cost, Quantity and Builder File Replacement %
 - Delta Cost – This delta is the cost difference between Actual Cost vs. Builder Cost

The table below is a simplified example of the above implementation steps

Work order											
Average					Builder File						
				Average Work	Replacement	Replacement					
Case	Component #	Part #	Quantity	Order Qty	%	%	Delta %	D/N	Actual Cost	Builder Cost	Delta Cost
1	1111	111-1111	2	1	50%	100%	50%	\$ 20	\$ 20	\$ 40	\$ 20
2	1111	222-2222	10	5	50%	50%	0%	\$ 10	\$ 50	\$ 50	\$ -
3	1111	333-3333	5	4	80%	20%	-60%	\$ 20	\$ 80	\$ 20	-\$ 60
									\$ 150	\$ 110	-\$ 40

Case 1 – The rebuild used less quantity of 111-1111 than Builder File's estimate. This is a favorable position for part number 111-1111. The localized Builder File can be adjusted for this part number. Lowering the cost of the rebuild would lower the TCO

Case 2 – Actual replacement rate of 222-2222 match with Builder File's estimate. Dealer can continue to use the Builder File replacement % in forecasting TCO

Case 3 – The rebuild used more quantity of 333-3333 than Builder File's estimate. This is not a favorable position for the part number as the actual repair consumed 60% more in quantity for this particular part number. This contributes \$60 more towards the TCO.

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The variance warrants further investigation and here are the areas to review:

- Confirm reusability guideline compliance during the rebuild process
 - Customer's operational and maintenance strategy could influence the reusability of certain part numbers. For example: higher than usual engine speed distribution can reduce the reusability of certain part numbers in components
 - Product Support Program is a mean for Caterpillar to address product issues in the field by providing commercial support to the customers. If the parts are replaced under such programs at rebuild then customer do not bear the cost. Hence it has a limited impact to the TCO and should be excluded from the analysis
 - When parts did not meet the reusability guideline and needed to be replaced at PCR, and these parts were not forecasted by the Builder File then SIMS tickets should be raised to inform Caterpillar of the higher than usually parts consumption and Caterpillar can address it through the CPI process. However, sometimes SIMS tickets might not be raised at the time of component rebuild and hence the parts consumption analysis is another mean to identify the discrepancy.
 - Furthermore, it is recommended Dealers' CRC to be involved to provide failed iron of the parts in question for further analysis and provide background why they were replaced
- **Accumulate Sufficient Samples and Repeat the Analysis** – To have reliable and representative data, a sufficient number of samples must be collected. The Dealer that assisted with the development of this Best Practice have over 100 rebuilds for certain LMT components.
 - **Bottom-up approach vs. Top-down approach** – The example above is a bottom-up approach whereby the analysis was done on an individual truck basis and then the data will be consolidated in an aggregate format. The advantage of this method is that it allows the analyst to drill down into individual truck level. However, as the machine population increases, the method might be time consuming to complete. An alternate method is to the top-down approach, whereby Dealers collect component rebuilds information for a large population and conduct the analysis in a macro level. The average replacement rate for a part piece is calculated based on the total number of the piece parts replaced divided the machine population. While this method is time efficient, it could overlook the effect of consist difference and serial number break. But based on previous experience the difference between to two methods would be within 5%, which is considered acceptable as we are trying to identify the variance of over 20% in the component analysis.

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4.0 Benefits

The benefits of the component analysis, particularly for mining machines, are:

- Understand impact of certain higher than usual part consumptions at rebuild have on the TCO and how TCO can be optimized
- Optimize parts consumption for rebuild and improve competitiveness in the rebuild market
- Enable further investigation into the root cause of the parts consumption variance between Builder File Estimate vs. Actual Parts Consumption for rebuild
- Provide feedback to Product Groups to initiate CPI projects to address certain part number(s) consumption that drive up TCO

5.0 Resources Required

- Management time to collect data, analyze, and review finding
- MS Excel or other analysis tool

6.0 Supporting Attachments / References

N/A

7.0 Related Best Practices

0608-4.3-1122 Forecasting CRC Component Rebuild Volume and Capacity
 0907-3.3-1086 Localizing a Centerline File for MARC Development
 0307-4.5-1068 Failure Analysis Process

8.0 Acknowledgements

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